

D. Ant on the Tree

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Connected undirected graph without cycles is called a tree. Trees is a class of graphs which is interesting not only for people, but for ants too.

An ant stands at the root of some tree. He sees that there are n vertexes in the tree, and they are connected by $n - 1$ edges so that there is a path between any pair of vertexes. A leaf is a distinct from root vertex, which is connected with exactly one other vertex.

The ant wants to visit every vertex in the tree and return to the root, passing every edge twice. In addition, he wants to visit the leaves in a specific order. You are to find some possible route of the ant.

Input

The first line contains integer n ($3 \leq n \leq 300$) — amount of vertexes in the tree. Next $n - 1$ lines describe edges. Each edge is described with two integers — indexes of vertexes which it connects. Each edge can be passed in any direction. Vertexes are numbered starting from 1. The root of the tree has number 1. The last line contains k integers, where k is amount of leaves in the tree. These numbers describe the order in which the leaves should be visited. It is guaranteed that each leaf appears in this order exactly once.

Output

If the required route doesn't exist, output -1 . Otherwise, output $2n - 1$ numbers, describing the route. Every time the ant comes to a vertex, output it's index.

Examples

input	Copy
3 1 2 2 3 3	
output	Copy
1 2 3 2 1	

input	Copy
6 1 2 1 3 2 4 4 5 4 6 5 6 3	
output	Copy
1 2 4 5 4 6 4 2 1 3 1	

input	Copy
6 1 2 1 3 2 4 4 5 4 6 5 3 6	
output	Copy
-1	

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 29 (Div. 2, Codeforces format)

Finished

Practice



→ Virtual participation

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Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2)

Choose file: No file chosen

→ Last submissions

Submission	Time	Verdict
373237816	May/01/2026 17:23	Accepted
373234083	May/01/2026 16:44	Accepted
373233685	May/01/2026 16:40	Wrong answer on test 2

→ **Problem tags**

constructive algorithms dfs and similar
trees *2000

No tag edit access

→ **Contest materials**

- Announcement 
- Tutorial (en) 

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Server time: May/01/2026 21:37:28^{UTC+7} (n2).

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